StudyBuddy

Team 10

Bhavya Trivedi, Vaughn Chan, Omar Fakousa, Sean Lee-Wah, Yash Bhupta, Aqsa Malik

Planning Document

Table of Contents

Table of Contents 1
Vision Statement 2
Big User Stories 3
Detailed User Stories 3
Customer meeting summary video 3

Vision Statement

StudyBuddy is a location-based platform that connects students who want to study together. Students create profiles with their courses and study preferences, then use an integrated map to discover nearby study buddies with the same course(s) and join or create study sessions at local libraries, cafes, and other study spots. The app combines social networking features with geographic discovery to make finding compatible study partners effortless.

The application will be useful for students who have troubles finding others in the same courses whom they can study with. For instance, some students prefer to study one-to-one, whereas others may feel safer studying in groups. The application offers a simple and safe way to meet with other students to gain both new peers and valuable knowledge about their course content.

The primary users of *StudyBuddy* are York University students who want to study together but are not able to find others to study with. This includes first-years, upper years, commuters, and study sessions. Firstly, this includes first-year students looking to build academic connections and mentors. Secondly, this includes upper-year students seeking focused, course-specific study groups. Thirdly, this includes commuter students who want location-based discovery and help preparing for midterms and finals. Lastly, this includes groups such as problem-solving sessions, clubs, and Peer Assisted Student Session (PASS), who can create new events to increase awareness about their program.

We will make the application only for York Students for security purposes. In addition, pseudo student users will be made to simulate the environment. Also, users should only be able to access York University meeting locations within the campus (i.e able to join Steacie's Library, but cannot meet up in York Village). In future releases, the application should integrate with York University's IT operations (i.e SIS integration) to properly verify the application.

We have a few options for authentication for restricting access to York University students. Firstly, two factor authentication through their email address being the primary option. In case the primary authentication method doesn't seem feasible, we can use Passport York's 'Duo' authentication. A further backup authentication method we could implement would be for users to scan their York ID and authenticate them. This would use OCR. Through these methods, *StudyBuddy* ensures a trusted, safe, and welcoming environment for our students and relevant user base while minimizing users' digital footprint, as well as privacy and safety concerns.

There are four key criteria that will define the success of the project if achieved. First, users are able to create and host event profiles for study sessions. Second, users are able to create student profiles and join or host study events through the application. Third, event profiles support a rating system that allows attendee users to provide feedback after each session. Finally, user and event profiles include keyword based highlights mentioned in their profiles which categorize their study preference and session type such as silent study, collaborative study or problem solving sessions etc.

In terms of market value, what makes our product unique is that it is specific to York University students (Added security with Passport York; less digital footprint), and has a more customizable matching algorithm. Additionally, it allows students to get matched with people taking the exact same course. (Not present in any existing solution). Students can also share materials and chat about course details. (current solutions do not have chat functionality). We encourage students to form a study session of 2-5 people, to encourage maximum participation and effective studying. (Current solutions do not have an adequate room size limit).

Big User Stories

As a student, I want to create a profile with my current courses and study preferences so that I can accurately be matched with compatible study partners.

Priority: high

Cost: 8 days

As a student, I want to make sure that I can trust the the person I study with and my information kept private

Priority: high

Cost: 7 days

As a student, I want to view an interactive map of active study sessions and nearby study spots so that I can easily find and join a group in my area.

Priority: high

Cost: 14 days

As a student, I want to be able to create study sessions and chat with potential attendees that is relevant to a specific topic, time, and location.

Priority: high

Cost: 13 days

Detailed User Stories

Student Profile

1. Due to JIRA counter reset restrictions ID 1.1.'s task was deleted in the planning process.

2. Create User Profile Page

Description: Display the user profile page. This shows information that includes name, email, major, bio, courses, study preferences, and past events attended (total meeting time and number of sessions).

3. View Study Session Log:

Description: Display a chronological history of all study sessions the student has previously attended or hosted. The log should include key details such as the date, time, location, topic covered, and a list of other participants, allowing the user to track their study habits over the semester.

Question: Should we limit the history to the current semester only to save storage?
Potentially display the length of time the user has attended study sessions for as well in session log to let others view how long they plan to work.
Priority: Medium
Cost: 3 days

4. Manage 'My Courses' List:

Description: Allow the student to add, remove, or update the specific courses they are currently enrolled in (Ex. EECS 2311, MATH 1013). This list acts as the primary filter for the matching algorithm, ensuring they only see relevant study partners and sessions for their current workload.

Priority: High
Cost: 2 days

5. Send Connection Request:

Description: Enable a user to send a friend request to another student they have successfully completed a session with. This action should trigger a notification to the recipient and, if accepted, add them to the user's 'My Network' list for easier future scheduling.

Priority: Medium
Cost: 4 days

6. Edit Study Preferences:

Description: Provide an interface for users to customize their 'Study Vibe' and behavioral preferences (Ex. 'Quiet Focus', 'Group Discussion', 'Whiteboard Work'). These tags are displayed on their public profile to help other students decide if they are a compatible match before joining a session.

Priority: High
Cost: 3 days

7. View Network List:

Description: specific page that displays a list of all accepted connections (friends). Users should be able to see the online status of their connections (Ex. 'Studying now at Scott Library') and have a quick-action button to invite them to a new session.

Priority: Low
Cost: 5 days

8. Customize Public Bio:

Description: Allow users to upload a profile picture and write a short text bio (max 200 characters) to introduce themselves. This helps build trust and familiarity, making other students feel more comfortable meeting up with a stranger for the first time.

Question: Do we need content moderation for the text bios?
Priority: Low
Cost: 2 days

Events

1. Create Events
As a user, I should be able to create an event profile for designated hosting.

Priority: High Cost: # day(s)

1.1. Host Events
As a user, I want to be able to select a place to host an event.

Priority: High Cost: # day(s)

2. Cancel Events
As a host of the event only I should be able to cancel the event.

Priority: High Cost: # day(s)

3. Join Existing and Upcoming Events
As a user, I want to be able to join one or more than one event a day.

Priority: High Cost: # day(s)

4. Cancel Events Attendance
As a user, I want to be able to cancel joining an event by myself only.

Priority: Medium Cost: ???

5. Privacy Policy - Events Specifics
As a user, I should be able to see all other attendees of the event that I joined to attend.

Priority: Medium Cost: # day(s)

6. Events Review
As an attending user, I should be able to review the event once the event finishes.

Priority: Medium Cost: # day(s)

7. Events Review Specifics
An event review should only be done by attending users of the event.

Priority: High Cost: # day(s)

8. Events Review Commenting
As a user, I should be able to see reviews of past events and be able to comment on these reviews to interact and initiate interest.

Priority: Medium Cost: # day(s)

Communication and Navigation:

1. View Possible Sessions

As a student, I want to be able to view available study sessions near me on a campus map, to easily find current groups near my current location

Priority: High Cost: # day(s)

2. View Details of Possible Sessions

As a student, I want to be able to view the details of study sessions appearing on the map so I can decide if I would like to join, based on these details (e.g. course, time, duration)

Priority: Medium Cost: # day(s)

3. Chat with Other Students

As a student, I want to be able to use a chat feature so students within a group can discuss things like study plans

Priority: High Cost: # day(s)

4. Add Attachments in Chat

As a student, I would like to be able to attach files and links in the chat so that I can communicate with my group about things like schedule conflicts

Priority: Medium Cost: # day(s)

5. Notifications for App

As a student, I would like to receive notifications for chat messages and session updates so that I do not miss important information while not viewing the app

Priority: Low Cost: # day(s)

6. Know Current Location

As a student, I would like to have my current location displayed on the map to make it easier for me to view nearby study sessions.

Priority: High Cost: # day(s)

7. Navigation to Study Location

As a student, I would like directional navigation on the map from my current location to the study location that is chosen so that I can get there efficiently.

Priority: High Cost: # day(s)

8. Online Status

As a student, I would like to be able to view the status of the members in my study group (online, idle, offline) so that it is easier to communicate with them knowing whether they are online or not.

Priority: Medium Cost: # day(s)

9. Indicator of Message

As a student, I want to be able to view an indicator when a message is currently being typed in the chat, to make it easier to communicate

Priority: Low Cost: # day(s)

Security & Privacy:

1. Information Privacy
Have explicit setting to reveal/hide certain information about the student

Priority: High Cost: # day(s)

2. 2FA Authentication
Have Email authentication to ensure that, even if people have my username or password, the intruder will not be able to login

Priority: High Cost: # day(s)

3. User Reports
Reporting system and secure database to keep track of users and reports, of which will only be available in our system

Priority: High Cost: # day(s)

4. Domain Filter
Only allow students with the @*my.yorku.ca* *or* @*yorku.ca* domain to sign up to the website

Priority: High Cost: # day(s)

5. Auto-Timeout
Auto-timeout after inactivity with the webpage, that of which is modifiable from 5 minutes to 30 minutes.

Priority: Medium Cost: # day(s)

Possible additional features for the future:

1. Precise Location Information: Users can choose in settings whether they want to share precise location information or just check whether they are in the general area of campus for safety purposes. This is for those that don't feel comfortable revealing their exact location.

2. Requesting Location Change: Confirmed/Interested participants to a StudyBuddy session should be allowed to send a request to the initiator of the session to request a change in venue/specific room which is subject to approval from them.

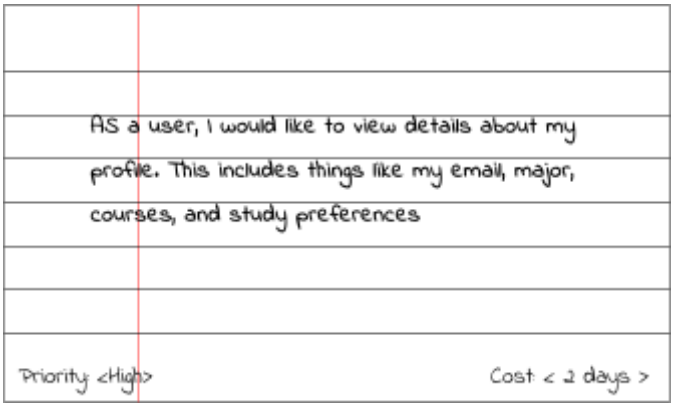
3. Easy In-App Coordination: Users need to be able to quickly communicate with each other last-minute when they reach a venue about where they're sitting and what they're wearing to be easily identifiable to each other for first time meet-ups.

4. Search Feature: Users search for study sessions, various filters will be available to narrow down the scope, such as: major, time, location.

Order of Detailed User Stories (top to bottom):

1. Student Profile
2. Events
3. Communication and Navigation
4. Security

<u>These were the detailed user stories that we were deciding to use for the ITR1 during the iteration. So, they may be different from our Jira board. However, the Jira board represents the most updated version of our task division for the development work. Please take a look at them if needed.</u>



AS a user, I would like to see the history of all sessions I have attended in the past, with details such as location and date.

Priority: <Medium> Cost: < 2 days >

As a user, I would like the ability to add and remove the specific courses I am currently taking. This would allow me to connect with relevant study partners.

Priority: High Cost: 2 days

As a user, would like to be able to send friend requests with students I have previously worked with. I would like them to receive a notification and if accepted, they would appear in a "My Network" list.

Priority: High Cost: 3 days

As a user, I would like to have a section that allows me to input my study preferences and display them on my profile. This would allow me to find compatible students to study with.

Priority: High Cost: 3 days

As a user, I would like to have a list that displays all of my connections. I would like to be able to see their online status, and be able to have a button to invite them to a new session.

Priority: Medium Cost: 4 Days

As a user, I would like to be able to customize my profile to build trust and familiarity with other students. This could include information like a bio and profile picture.

Priority: High Cost: 2 days

As a user, I would like to be able to use an event profile to be able to host an event for others to join.

Priority: High Cost: 2 days

As a user, I want to be able to join one or more than one event a day.

Priority: High Cost: 1 day

As a user, I want to be able to cancel joining an event, by myself only.

Priority: High

Cost: 1 day

As a user, I should be able to see all other attendees of the event that I joined.

Priority: High

Cost: 2 days

As an attending user, I should be able to review the details of the event once it finishes.

Priority: Medium

Cost: 1 day

As a user, I would like to be able to create events and an event profile to allow me to join events.

Priority: High

Cost: 3 days

As a user, it should be mandatory that an event review can only be done by attending users of that specific event.

Priority: High

Cost: 1 day

As a user, I should be able to see reviews of past events and be able to comment on these reviews to interact and initiate interest with others.

Priority: Medium

Cost: 2 days

As a user, I want to be able to view available study sessions near me on a campus map, to easily find current groups near my current location.

Priority: High | Cost: 3 days

As a user, I want to be able to view the details of study sessions appearing on the map so I can decide if I would like to join, based on these details (e.g. course, time, duration).

Priority: High | Cost: 3 days

As a user, I want to be able to use a chat feature so students within my group can discuss things like study plans.

Priority: High | Cost: 2 days

As a user, I would like to be able to attach files and links in the chat so that I can communicate with my group about things like schedule conflicts.

Priority: Low | Cost: 1 day

As a user, I would like to receive notifications for chat messages and session updates so that I do not miss important information while not viewing the app.

Priority: Low | Cost: 1 day

As a user, I would like to have my current location displayed on the map to make it easier for me to view nearby study sessions.

Priority: High | Cost: 1 day

As a user, I would like to have directional navigation on the map from my current location to the study location that is chosen so that I can get there efficiently.

Priority: Medium | Cost: 3 days

As a user, I would like to be able to view the status of the members in my study group (online, idle, offline) so that it is easier to communicate with them knowing whether they are online or not.

Priority: Medium | Cost: 1 day

As a user, I want to be able to view an indicator when a message is currently being typed in the chat, to make it easier to communicate.

Priority Low | Cost 1 day

As a user, I want to have explicit settings about whether I want to reveal certain information about myself.

Priority High | Cost 2 days

As a user, I want to have email authentication to ensure that even if someone has my username and password, they will not be able to login to my account.

Priority High | Cost 2 days

As a user, I want the app to have a secure database to keep track of both users and reports. This would ensure my data is safe from others.

Priority High | Cost 3 days

As a user, I want the app to be restricted to only people from York University. This means only accepting emails ending with @my.yorku.ca and @yorku.ca when signing up.

Priority High | Cost 2 days

As a user, I want there to be an auto-timeout feature enabled in the app after a certain amount time.

Priority Low | Cost 2 days

Iteration Specific Development Plan

1. Executive Summary

This project follows a **Vertical Slice** strategy. We will build the application in three iterations (MVP, Community, Experience), ensuring that the Backend (Logic/DB), Frontend (View), and Map components progress in parallel. The architecture follows the **MVC (Model-View-Controller)** pattern defined in our planning session.

2. Strategic Roadmap (3 Iterations)

Iteration 1: The "Connector" (MVP)

Goal: A student can log in safely, define their courses, and find a physical location to study.

- Vaughn - **Architecture:** Setup the MVC directory structure (Model for DB logic, View for UI, Controller for routing).
- Aqsa - **Auth (Security):** Implement Domain Filter (`@my.yorku.ca` only).
- Omar - **Profile:** "My Courses" List. Create the schema for `Student` (Name, Email, Major, Courses[]) as per the whiteboard schema.
- Bhavya - **Events/Nav:** "Home Event Page" Dashboard.
- **Sean - Events/Nav:** Base Map View. Show pins for active sessions.
- **Yash - Action:** Create a Session (The "Drop a Pin" feature).

Iteration 2: The "Community" (Working Version)

Goal: Students can communicate, save interest in events, and filter for specific study vibes.

- **Communication:** Chat System. (One-on-one or Group chat for a specific Event).
- **Events:** Wishlist/Saves. Allow users to "Star" an event they are interested in (from Diagram 1).
- **Profile:** Study Preferences & Vibe Tags. Implement the `Study Preferences*` field from the schema (Quiet vs. Group).
- **Search:** Advanced Filtering. Filter map pins by "Major," "Course Code," or "Time."
- **Network:** Connection Requests. Allow users to add "Study Buddies" (populating the `List of Connections[]` array).

Iteration 3: The "Experience" (Polished Prototype)

Goal: Retention features, gamification, and advanced visuals.

- **Gamification:** "Study Streaks" & Challenges. Display hours studied or consecutive days (from Diagram 1).
- **Smart Features:** AI-Generated Tips or Notifications based on upcoming tests.
- **Network:** Visual Graph. Display the "You -> Friend -> Friend" node graph (from Diagram 1) to show network reach.
- **History:** Past Events Log. Populating the `Past Events Attended[]` schema for user review.
- **Security:** User Reporting & Block functionality.

3. Master Task List (Jira/Trello Ready)

Team Size: 6 Members **Role Key:**

- **BE:** Backend/Database (Members 1 & 2)
- **FE:** Frontend/UI (Members 3 & 4)
- **Full Stack/Map:** Map Integration & Logic (Members 5 & 6)

Customer meeting summary video:

Please see attached video: 🎬 Customer meeting's summary video.mp4